

Tensor Time Series Imputation through Tensor Factor Modelling

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Abstract

We propose tensor time series imputation when the missing pattern in the tensor data can be general, as long as any two data positions along a tensor fibre are both observed for enough time points. The method is based on a tensor time series factor model with Tucker decomposition of the common component. One distinguished feature of the tensor time series factor model used is that there can be weak factors in the factor loadings matrix for each mode. This reflects reality better when real data can have weak factors which drive only groups of observed variables, for instance, a sector factor in financial market driving only stocks in a particular sector. Using the data with missing entries, asymptotic normality is derived for rows of estimated factor loadings, while consistent covariance matrix estimation enables us to carry out inferences. As a first in the literature, we also propose a ratio-based estimator for the rank of the core tensor under general missing patterns. Rates of convergence are spelt out for the imputations from the estimated tensor factor models. We introduce a new measure for gauging imputation performances, and simulation results show that our imputation procedure works well, with asymptotic normality and corresponding inferences also demonstrated. Re-imputation performances are also gauged when we demonstrate that using slightly larger rank than estimated gives superior re-imputation performances. An NYC taxi traffic data set is also analyzed by imposing general missing patterns and gauging the imputation performances.

Key words and phrases: Generalized cross-covariance matrix, tensor unfolding, core tensor, α -mixing time series variables, missingness tensor.

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1 Introduction

Large dimensional panel data is easier to obtain than ever thanks to a quickly evolving internet speed and more diverse download platforms. Together with the advancement of statistical analyses for these data over the past decade, researchers also open up more to time series data with higher order, namely, tensor time series data. A prominent example would be order 2 tensor time series, i.e., matrix-valued time series. Wang et al. (2019) proposes a factor model using a Tucker decomposition of the common component in the modelling. An example on monthly import-export volume of products among different countries is given in Chen et al. (2022a), where factor modelling using Tucker decomposition is explored, and generalized to higher order tensors. Focusing on matrix-valued time series, Chang et al. (2023) proposes a tensor-CP decomposition for modelling the data. Zhang (2024) and Chen et al. (2021) propose autoregressive models for matrix-valued time series. For a more comprehensive review on matrix-valued time series analysis, please refer to Tsay (2023).

A less addressed topic in large time series analysis is the treatment of missing data, in particular, the imputation of missing data and corresponding inferences. While there are numerous data-centric methods in various scientific fields for imputing multivariate time series data (see Chapon et al. (2023) for environmental time series, Kazijevs and Samad (2023) for health time series, Zhao et al. (2023) and Zhang et al. (2021) for using deep-learning related architectures for imputations, to name but a few), almost none of them address statistically how accurate their methods are, and all of them are not for higher order tensor time series. We certainly can line up the variables in a tensor time series to make it a longitudinal panel, but in doing so we lose special structures and insights that can be utilized for forecasting and interpretation of the data. More importantly, transforming a moderate sized tensor to a vector means the length of the vector can be much larger than the sample size, creating curse of dimensionality.

For imputing large panel of time series with statistical analyses, Bai and Ng (2021) defines the concept of TALL and WIDE blocks of data and proposes an iterative TW algorithm in imputing missing values in a large panel, while Cahan et al. (2023) improves the TW algorithm to a Tall-Project (TP) algorithm so that there is no iterations needed. Both papers use factor modelling for the imputations, and derive rates of convergence when all factors are pervasive and the number of factors known. Asymptotic normality for rows of estimated factor loadings and the corresponding practical inferences are also developed as well. Xiong and Pelger (2023) also bases their imputations on a factor model for a large panel of time series with pervasive factors and number of factors known, and build a method for imputing missing values under very general missing patterns, with asymptotic normality and inferences also developed.

To the best of our knowledge, for tensor time series with order larger than 1 (i.e, at least matrix-valued), there are no theoretical analyses on imputation performances. Imputation methodologies developed on

tensor time series are also scattered around very different applications. See Chen et al. (2022b) on traffic tensor data and Pan et al. (2021) for RNA-sequence tensor data for instance.

In view of all the above, as a first in the literature, we aim to develop a tensor imputation method accompanied by theoretical analyses in this paper. Like Cahan et al. (2023), we use factor modelling for tensor time series as a basis for our imputation method. Unlike Cahan et al. (2023), Bai and Ng (2021) or Xiong and Pelger (2023) though, we develop a method that can consistently estimate the number of factors, or the core tensor rank, in a Tucker decomposition-based factor model for the tensor time series with missing values. On top of this, we also allow factors to be weak. A weak factor corresponds to a column in a factor loading matrix being sparse, or approximately sparse. This implies that not all units in a tensor has dynamics contributed by all the factors inside the core tensor. In Chen and Lam (2024), they allow for weak factors in their analyses, and discovers that there are potentially weak factors in the NYC taxi traffic data, which we are going to analyse in Section 5.2.2. We prove consistency of our imputations under general missingness, and develop asymptotic normality and practical inferences for rows of factor loading matrix estimators, with rates of convergence in all consistency results spelt out. Our method is available in the R package `tensorMiss`, which has used the `Rcpp` package to greatly boost computational speed.

The rest of the paper is organised as follows. Section 2 introduces the notations used in this paper. Section 3 describe the tensor factor model and the imputation methodology we use. Section 4 lays down the main assumptions for the paper, with consistent estimation and rates of convergence of all factor loading matrix estimators and imputed values presented. Asymptotic normality and the estimators of the corresponding covariance matrices for practical inferences are introduced as well in Section 4.3, before our proposed ratio-based estimators for the number of factors in Section 4.5. Section 5 presents extensive simulation results for our paper, together with an analysis for the NYC taxi traffic data in Section 5.2.2. All proofs are in the supplementary materials associated with this paper.

2 Notations

Throughout this paper, we use the lower-case letter, bold lower-case letter, bold capital letter, and calligraphic letter, i.e. a , $\mathbf{a}$, $\mathbf{A}$, $\mathcal{A}$, to denote a scalar, a vector, a matrix, and a tensor. We also use a_i , A_{ij} , $\mathbf{A}_i$, $\mathbf{A}_{\cdot i}$ to denote, respectively, the i -th element of $\mathbf{a}$, the (i, j) -th element of $\mathbf{A}$, the i -th row vector (as a column vector) of $\mathbf{A}$, and the i -th column vector of $\mathbf{A}$. We use $\otimes$ to represent the Kronecker product, and $\circ$ the Hadamard product. We use $a \asymp b$ to denote $a = O(b)$ and $b = O(a)$.

Some tensor-related notations. Hereafter, given a positive integer m , define $[m] := \{1, 2, \dots, m\}$. For an order- K tensor $\mathcal{A} = (A_{i_1, \dots, i_K}) \in \mathbb{R}^{d_1 \times \dots \times d_K}$, a column vector $(A_{i_1, \dots, i_{k-1}, j, i_{k+1}, \dots, i_K})_{j \in [d_k]}$ represents

a mode- k fibre for the tensor $\mathcal{A}$. We denote by $\text{mat}_k(\mathcal{A}) \in \mathbb{R}^{d_k \times d_{-k}}$ (or sometimes $\mathbf{A}_{(k)}$, with $d_{-k} := (\prod_{j=1}^K d_j)/d_k$) the mode- k unfolding of a tensor, defined by placing all mode- k fibres into a matrix. We denote by $\mathcal{M} \times_k \mathbf{A}$ the mode- k multiplication of a tensor $\mathcal{M}$ with a matrix $\mathbf{A}$, defined by

$$\text{mat}_k(\mathcal{M} \times_k \mathbf{A}) := \mathbf{A} \text{mat}_k(\mathcal{M}).$$

The notation $\text{vec}(\cdot)$ denotes the vectorisation of a matrix or the vectorisation of the mode-1 unfolding of a tensor. The i -th largest eigenvalue of a matrix $\mathbf{A}$ is denoted by $\lambda_i(\mathbf{A})$. The notation $\mathbf{A} \succcurlyeq 0$ (resp. $\mathbf{A} \succ 0$) means that $\mathbf{A}$ is positive semi-definite (resp. positive definite). We use $\mathbf{A}'$ to denote the transpose of $\mathbf{A}$. We use $\text{diag}(\mathbf{A})$ to denote a diagonal matrix with the diagonal elements of $\mathbf{A}$, and $\text{diag}(\{a_1, \dots, a_n\})$ represents the diagonal matrix with $\{a_1, \dots, a_n\}$ on the diagonal.

Some norm notations. For a given set, we denote by $|\cdot|$ its cardinality. We use $\|\cdot\|$ to denote the spectral norm of a matrix or the L_2 norm of a vector, and $\|\cdot\|_F$ to denote the Frobenius norm of a matrix. We use $\|\cdot\|_{\max}$ to denote the maximum absolute value of the elements in a vector, a matrix or a tensor. The notations $\|\cdot\|_1$ and $\|\cdot\|_{\infty}$ denote the L_1 and L_{∞} -norm of a matrix respectively, defined by $\|\mathbf{A}\|_1 := \max_j \sum_i |(\mathbf{A})_{ij}|$ and $\|\mathbf{A}\|_{\infty} := \max_i \sum_j |(\mathbf{A})_{ij}|$. WLOG, we always assume the eigenvalues of a matrix are arranged by descending orders, and so are their corresponding eigenvectors. For more details on tensor manipulations, readers are referred to Kolda and Bader (2009).

3 Tensor Factor Model with Missing Entries

Suppose we observe an order- K mean-zero tensor $\mathcal{Y}_t = (\mathcal{Y}_{t,i_1,\dots,i_K}) \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_K}$ for each $t \in [T]$, modelled by

$$\mathcal{Y}_t = \mathcal{C}_t + \mathcal{E}_t = \mathcal{F}_t \times_1 \mathbf{A}_1 \times_2 \mathbf{A}_2 \times_3 \dots \times_K \mathbf{A}_K + \mathcal{E}_t, \quad t \in [T], \quad (3.1)$$

where $\mathcal{C}_t$ is the common component and $\mathcal{E}_t$ the error tensor. The core tensor is $\mathcal{F}_t \in \mathbb{R}^{r_1 \times r_2 \times \dots \times r_K}$, and each mode- k factor loading matrix $\mathbf{A}_k$ has dimension $d_k \times r_k$. We define $d := \prod_{k=1}^K d_k$, $r := \prod_{k=1}^K r_k$, and also $d_{-k} := d/d_k$, and $r_{-k} := r/r_k$. The above model is an extension to the usual time series factor model ($K = 1$):

$$\mathcal{Y}_t = \text{mat}_1(\mathcal{Y}_t) = \text{mat}_1(\mathcal{F}_t \times_1 \mathbf{A}_1) + \text{mat}_1(\mathcal{E}_t) = \mathbf{A}_1 \text{mat}_1(\mathcal{F}_t) + \text{mat}_1(\mathcal{E}_t) = \mathbf{A}_1 \mathcal{F}_t + \mathcal{E}_t,$$

and also for a matrix-valued time series factor model ($K = 2$):

$$\mathcal{Y}_t = \text{mat}_1(\mathcal{Y}_t) = \mathbf{A}_1 \text{mat}_1(\mathcal{F}_t) \mathbf{A}_2' + \text{mat}_1(\mathcal{E}_t) = \mathbf{A}_1 \mathcal{F}_t \mathbf{A}_2' + \mathcal{E}_t.$$

We normalise each factor loading matrix to $\mathbf{Q}_k = \mathbf{A}_k \mathbf{Z}_k^{-1/2}$, where $\mathbf{Z}_k = \text{diag}(\mathbf{A}_k' \mathbf{A}_k)$ contains the square norm of each column of $\mathbf{A}_k$, representing the strength of each factor. For instance, a pervasive factor (so that the corresponding column in $\mathbf{A}_k$ is dense) has order d_k in the corresponding position in $\mathbf{Z}_k$, while a weak factor (a sparser column in $\mathbf{A}_k$) can have order d_k^α for some $\alpha < 1$. We allow the factors to have different strength, so that $\mathbf{Z}_k$ can have values of different orders. Such a model is closer to reality, for instance apart from a pervasive market factor, there can be weaker sector factors in a large selection of stock returns. For the exact technical details, see Assumption (L1) in Section 4.1.

With the normalization above, we can then write the mode- k unfolding of the data as:

$$\begin{aligned} \text{mat}_k(\mathcal{Y}_t) &= \text{mat}_k(\mathcal{F}_t \times_1 \mathbf{A}_1 \times_2 \mathbf{A}_2 \times_3 \cdots \times_K \mathbf{A}_K) + \text{mat}_k(\mathcal{E}_t) \\ &= \mathbf{Q}_k \text{mat}_k(\mathcal{F}_{Z,t}) \mathbf{\Lambda}_k' + \text{mat}_k(\mathcal{E}_t), \end{aligned} \quad (3.2)$$

where $\mathcal{F}_{Z,t} := \mathcal{F}_t \times_1 \mathbf{Z}_1^{1/2} \times_2 \cdots \times_K \mathbf{Z}_K^{1/2}$, and $\mathbf{\Lambda}_k := \otimes_{j \in [K] \setminus \{k\}} \mathbf{Q}_j$ with the index of product being descending.

We only observe partial data. Based on the missing pattern, we define the missingness tensor $\mathcal{M}_t = (\mathcal{M}_{t,i_1,\dots,i_K}) \in \mathbb{R}^{d_1 \times d_2 \times \cdots \times d_K}$ with

$$\mathcal{M}_{t,i_1,\dots,i_K} = \begin{cases} 1, & \text{if } \mathcal{Y}_{t,i_1,\dots,i_K} \text{ is observed;} \\ 0, & \text{otherwise.} \end{cases}$$

In this paper, we make use of the following notation:

$$\psi_{k,ij,h} := \left\{ t \in [T] \mid \text{mat}_k(\mathcal{M}_t)_{ih} \text{mat}_k(\mathcal{M}_t)_{jh} = 1 \right\}. \quad (3.3)$$

Hence $\psi_{k,ij,h}$ is the set of time periods where both the i -th and j -th entries of the h -th mode- k fibre are observed, where $i, j \in [d_k]$, $h \in [d_{-k}]$. Our aim is to recover the value for the common component $\mathcal{C}_{t,i_1,\dots,i_K}$ if $\mathcal{M}_{t,i_1,\dots,i_K} = 0$. Assuming first the number of factors r_k is known for all modes, we want to obtain the estimators of the factor loading matrices, $\widehat{\mathbf{Q}}_k$ for $k \in [K]$, and then the estimated core tensor series $\widehat{\mathcal{F}}_{Z,t}$ for $t \in [T]$. Imputation then follows by $\widehat{\mathcal{C}}_t := \widehat{\mathcal{F}}_{Z,t} \times_1 \widehat{\mathbf{Q}}_1 \times_2 \cdots \times_K \widehat{\mathbf{Q}}_K$. We leave the discussion of estimating r_k to Section 4.5.

3.1 Estimation of factor loading matrices

Our method relies on the reconstruction of the mode- k sample covariance matrix $\mathbf{S}_k$, defined for $i, j \in [d_k]$,

$$(S_k)_{ij} := \frac{1}{T} \sum_{t=1}^T \text{mat}_k(\mathcal{Y}_t)'_{i\cdot} \text{mat}_k(\mathcal{Y}_t)_{j\cdot} = \sum_{h=1}^{d_k} \frac{1}{T} \sum_{t=1}^T \text{mat}_k(\mathcal{Y}_t)_{ih} \text{mat}_k(\mathcal{Y}_t)_{jh}. \quad (3.4)$$

With missing entries characterized by $\mathcal{M}_t$ and $\psi_{k,ij,h}$ in (3.3), we can generalise the above to

$$(\hat{S}_k)_{ij} = \sum_{h=1}^{d_k} \left\{ \frac{1}{|\psi_{k,ij,h}|} \sum_{t \in \psi_{k,ij,h}} \text{mat}_k(\mathcal{Y}_t)_{ih} \text{mat}_k(\mathcal{Y}_t)_{jh} \right\}. \quad (3.5)$$

Intuitively, the cross-covariance between unit i and j at the h -th mode- k fibre is estimated inside the curly bracket in (3.5) using only the corresponding available data. PCA can now be performed on $\hat{\mathbf{S}}_k$, and $\hat{\mathbf{Q}}_k$ is obtained as the first r_k eigenvectors of $\hat{\mathbf{S}}_k$.

3.2 Imputation and estimation of the core tensor series

With $\hat{\mathbf{Q}}_k$ available (which is estimating the factor loading space of $\mathbf{Q}_k$, albeit $\mathbf{Q}_k$ does not contain orthogonal columns, but $\hat{\mathbf{Q}}_k$ does), we can estimate $\mathcal{F}_{Z,t}$ (equivalently $\text{vec}(\mathcal{F}_{Z,t})$) by observing that

$$\text{vec}(\mathcal{Y}_t) = \mathbf{Q}_{\otimes} \text{vec}(\mathcal{F}_{Z,t}) + \text{vec}(\mathcal{E}_t), \quad \text{where } \mathbf{Q}_{\otimes} := \mathbf{Q}_K \otimes \cdots \otimes \mathbf{Q}_1.$$

If $\mathbf{Q}_{\otimes}$ is known, then the least squares estimator of $\text{vec}(\mathcal{F}_{Z,t})$ is given by

$$\text{vec}(\mathcal{F}_{Z,t}) = (\mathbf{Q}'_{\otimes} \mathbf{Q}_{\otimes})^{-1} \mathbf{Q}'_{\otimes} \text{vec}(\mathcal{Y}_t) = \left(\sum_{j=1}^d \mathbf{Q}_{\otimes,j\cdot} \mathbf{Q}'_{\otimes,j\cdot} \right)^{-1} \left(\sum_{j=1}^d \mathbf{Q}_{\otimes,j\cdot} [\text{vec}(\mathcal{Y}_t)]_j \right).$$

With missing data, using the missingness tensor $\mathcal{M}_t$, the above can be generalized to

$$\text{vec}(\hat{\mathcal{F}}_{Z,t}) = \left(\sum_{j=1}^d [\text{vec}(\mathcal{M}_t)]_j \hat{\mathbf{Q}}_{\otimes,j\cdot} \hat{\mathbf{Q}}'_{\otimes,j\cdot} \right)^{-1} \left(\sum_{j=1}^d [\text{vec}(\mathcal{M}_t)]_j \hat{\mathbf{Q}}_{\otimes,j\cdot} [\text{vec}(\mathcal{Y}_t)]_j \right). \quad (3.6)$$

We can then estimate the common components at time t by

$$\hat{\mathcal{C}}_t = \hat{\mathcal{F}}_{Z,t} \times_1 \hat{\mathbf{Q}}_1 \times_2 \cdots \times_K \hat{\mathbf{Q}}_K. \quad (3.7)$$

With (3.7), we can impute $\mathcal{Y}_t$ using

$$\tilde{\mathcal{Y}}_{t,i_1,\dots,i_K} = \begin{cases} \mathcal{Y}_{t,i_1,\dots,i_K}, & \text{if } \mathcal{M}_{t,i_1,\dots,i_K} = 1; \\ \hat{\mathcal{C}}_{t,i_1,\dots,i_K}, & \text{if } \mathcal{M}_{t,i_1,\dots,i_K} = 0. \end{cases}$$

4 Assumptions and Theoretical Results

We present our assumptions for consistent imputation and estimation of factor loading matrices, with the corresponding theoretical results presented afterwards.

4.1 Assumptions

(O1) (Observation patterns)

1. $\mathcal{M}_t$ is independent of $\mathcal{F}_s$ and $\mathcal{E}_s$ for any $t, s \in [T]$.
2. Given $\mathcal{M}_t$ with $t \in [T]$, for any $k \in [K], i, j \in [d_k], h \in [d_{-k}]$, there exists a constant ψ_0 such that

$$\frac{|\psi_{k,ij,h}|}{T} \geq \psi_0 > 0.$$

(M1) (Alpha mixing) *The elements in $\mathcal{F}_t$ and $\mathcal{E}_t$ are α -mixing. A vector process $\{\mathbf{x}_t : t = 0, \pm 1, \pm 2, \dots\}$ is α -mixing if, for some $\gamma > 2$, the mixing coefficients satisfy the condition that*

$$\sum_{h=1}^{\infty} \alpha(h)^{1-2/\gamma} < \infty,$$

where $\alpha(h) = \sup_{\tau} \sup_{A \in \mathcal{H}_{-\infty}^{\tau}, B \in \mathcal{H}_{\tau+h}^{\infty}} |\mathbb{P}(A \cap B) - \mathbb{P}(A)\mathbb{P}(B)|$ and $\mathcal{H}_{\tau}^s$ is the σ -field generated by $\{\mathbf{x}_t : \tau \leq t \leq s\}$.

(F1) (Time series in $\mathcal{F}_t$) *There is $\mathcal{X}_{f,t}$ the same dimension as $\mathcal{F}_t$, such that $\mathcal{F}_t = \sum_{q \geq 0} a_{f,q} \mathcal{X}_{f,t-q}$. The time series $\{\mathcal{X}_{f,t}\}$ has i.i.d. elements with mean 0 and variance 1, with uniformly bounded fourth order moments. The coefficients $a_{f,q}$ are such that $\sum_{q \geq 0} a_{f,q}^2 = 1$ and $\sum_{q \geq 0} |a_{f,q}| \leq c$ for some constant c .*

(L1) (Factor strength) *We assume for $k \in [K]$, $\mathbf{A}_k$ is of full rank and independent of factors and errors series. Furthermore, as $d_k \rightarrow \infty$,*

$$\mathbf{Z}_k^{-1/2} \mathbf{A}_k' \mathbf{A}_k \mathbf{Z}_k^{-1/2} \rightarrow \Sigma_{A,k}, \quad (4.1)$$

where $\mathbf{Z}_k = \text{diag}(\mathbf{A}_k' \mathbf{A}_k)$ and $\Sigma_{A,k}$ is positive definite with all eigenvalues bounded away from 0 and

infinity. We assume $(\mathbf{Z}_k)_{jj} \asymp d_k^{\alpha_{k,j}}$ for $j \in [r_k]$, and $1/2 < \alpha_{k,r_k} \leq \dots \leq \alpha_{k,2} \leq \alpha_{k,1} \leq 1$.

With Assumption (L1), we can denote $\mathbf{Q}_k := \mathbf{A}_k \mathbf{Z}_k^{-1/2}$ and hence $\mathbf{Q}'_k \mathbf{Q}_k \rightarrow \boldsymbol{\Sigma}_{A,k}$. We need $\alpha_{k,j} > 1/2$ in order that the ratio-based estimator of the number of factors in Section 4.5 works.

(E1) (Decomposition of $\mathcal{E}_t$) We assume K is constant, and

$$\mathcal{E}_t = \mathcal{F}_{e,t} \times_1 \mathbf{A}_{e,1} \times_2 \dots \times_K \mathbf{A}_{e,K} + \boldsymbol{\Sigma}_\epsilon \circ \boldsymbol{\epsilon}_t, \quad (4.2)$$

where $\mathcal{F}_{e,t}$ is an order- K tensor with dimension $r_{e,1} \times \dots \times r_{e,K}$, containing independent elements with mean 0 and variance 1. The order- K tensor $\boldsymbol{\epsilon}_t \in \mathbb{R}^{d_1 \times \dots \times d_K}$ contains independent mean zero elements with unit variance, with the two time series $\{\boldsymbol{\epsilon}_t\}$ and $\{\mathcal{F}_{e,t}\}$ being independent. The order- K tensor $\boldsymbol{\Sigma}_\epsilon$ contains the standard deviations of the corresponding elements in $\boldsymbol{\epsilon}_t$, and has elements uniformly bounded.

Moreover, for each $k \in [K]$, $A_{e,k} \in \mathbb{R}^{d_k \times r_{e,k}}$ is such that $\|\mathbf{A}_{e,k}\|_1 = O(1)$. That is, $\mathbf{A}_{e,k}$ is (approximately) sparse.

(E2) (Time series in $\mathcal{E}_t$) There is $\mathcal{X}_{e,t}$ the same dimension as $\mathcal{F}_{e,t}$, and $\mathcal{X}_{\epsilon,t}$ the same dimension as $\boldsymbol{\epsilon}_t$, such that $\mathcal{F}_{e,t} = \sum_{q \geq 0} a_{e,q} \mathcal{X}_{e,t-q}$ and $\boldsymbol{\epsilon}_t = \sum_{q \geq 0} a_{\epsilon,q} \mathcal{X}_{\epsilon,t-q}$, with $\{\mathcal{X}_{e,t}\}$ and $\{\mathcal{X}_{\epsilon,t}\}$ independent of each other, and each time series has independent elements with mean 0 and variance 1 with uniformly bounded fourth order moments. Both $\{\mathcal{X}_{e,t}\}$ and $\{\mathcal{X}_{\epsilon,t}\}$ are independent of $\{\mathcal{X}_{f,t}\}$ from (F1).

The coefficients $a_{e,q}$ and $a_{\epsilon,q}$ are such that $\sum_{q \geq 0} a_{e,q}^2 = \sum_{q \geq 0} a_{\epsilon,q}^2 = 1$ and $\sum_{q \geq 0} |a_{e,q}|, \sum_{q \geq 0} |a_{\epsilon,q}| \leq c$ for some constant c .

(R1) (Further rate assumptions) We assume that, with $g_s := \prod_{k=1}^K d_k^{\alpha_{k,1}}$,

$$dg_s^{-2} T^{-1} d_k^{2(\alpha_{k,1} - \alpha_{k,r_k}) + 1} = o(1), \quad dg_s^{-1} T^{-1} d_k^{2(\alpha_{k,1} - \alpha_{k,r_k})} = o(1), \quad dg_s^{-1} d_k^{\alpha_{k,1} - \alpha_{k,r_k} - 1/2} = o(1).$$

Assumption (O1) means that the missing mechanism is independent of the factors and the noise series, which is also assumed in Xiong and Pelger (2023) for the purpose of identification. It also means that the missing pattern can depend on the K factor loading matrices, allowing for a wide variety of missing patterns that can vary over time and units in different dimensions. Condition 2 of (O1) implies that the number of time periods that any two individual units are both observed are at least proportional to T , which simplifies proofs and presentations, and is also used in Xiong and Pelger (2023). Assumption (M1) is a standard assumption in vector time series factor models, which facilitates proofs using central limit theorem for time series without losing too much generality. Assumption (F1), (E1) and (E2) are

exactly the corresponding assumptions in Chen and Lam (2024), allowing for serial correlations in the factor series, and serial and cross-sectional dependence within and among the error tensor fibres. These three assumptions facilitate the proof of asymptotic normality in Section 4.3. Together with Assumption (M1), we implicitly restrict the general linear processes in (F1) and (E2) to be, for instance, of short rather than long dependence.

Assumption (L1) is quite different from assumptions in other papers on factor models, in the sense that we allow for the existence of weak factors alongside the pervasive ones. Chen and Lam (2024) adapted the same assumption, which allows each column of $\mathbf{A}_k$ to be completely dense (i.e., a pervasive factor) or sparse to a certain extent. A diagonal entry in $\mathbf{Z}_k$ then records how dense a column really is, and the corresponding strength of factors defined.

Finally, Assumption (R1) are the technical rate assumptions needed for the proof of various theorems in the paper because of the existence of weak factors. If all factors are pervasive (i.e., $\alpha_{k,j} = 1$), then the conditions are automatically satisfied. Suppose $K = 2$, $T \asymp d_1 \asymp d_2$ and the strongest factors are all pervasive (i.e., $\alpha_{k,1} = 1$), then we need $\alpha_{k,r_k} > 1/2$ for (R1) to be satisfied. This condition is the same as the one remarked right after we stated Assumption (L1). A factor with $\alpha_{k,j}$ close to 0.5 presents a significantly weak factor with only more than $d_k^{1/2}$ of elements are non-zero in the corresponding column of $\mathbf{A}_k$.

Remark 1 *With the missing entries imputed using the estimated common components $\widehat{\mathcal{C}}_{t,i_1,\dots,i_K}$, we have a completed data set which could be used for re-estimation and hence re-imputation. The convergence could be shown empirically to be accelerated by such a procedure. The rate improvement would be from the difference between T and $\psi_0 T$, where ψ_0 is the lowest proportion of observation among all entries from Assumption (O1). We omit the lengthy proofs as eventually the rates only differ by a constant, but we note here that re-imputation can indeed improve our imputation, which is essentially credited to the more observations used when we have an initially good imputation.*

4.2 Consistency: factor loadings and imputed values

We first present a consistency result for the factor loading matrix estimator $\widehat{\mathbf{Q}}_k$ of $\mathbf{Q}_k$. For $k \in [K]$, $j \in [d_k]$, define

$$\mathbf{H}_{k,j} := \widehat{\mathbf{D}}_k^{-1} \sum_{i=1}^{d_k} \widehat{\mathbf{Q}}_{k,i} \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{k,ij,h}|} \sum_{t \in \psi_{k,ij,h}} \left(\sum_{m=1}^{r_k} \Lambda_{k,hm} \text{mat}_k(\mathcal{F}_{Z,t})_{\cdot m} \right)' \mathbf{Q}_{k,i} \cdot \left(\sum_{m=1}^{r_k} \Lambda_{k,hm} \text{mat}_k(\mathcal{F}_{Z,t})_{\cdot m} \right)', \quad (4.3)$$

$$\mathbf{H}_k^a := \frac{1}{T} \sum_{t=1}^T \widehat{\mathbf{D}}_k^{-1} \widehat{\mathbf{Q}}_k' \mathbf{Q}_k \text{mat}_k(\mathcal{F}_{Z,t}) \Lambda_k' \Lambda_k \text{mat}_k(\mathcal{F}_{Z,t})', \quad (4.4)$$

where $\widehat{\mathbf{D}}_k := \widehat{\mathbf{Q}}_k' \widehat{\mathbf{S}}_k \widehat{\mathbf{Q}}_k$ is a diagonal matrix of eigenvalues of $\widehat{\mathbf{S}}_k$ defined in (3.5). Hence $\mathbf{H}_{k,j} = \mathbf{H}_k^a$ if there are no missing entries, i.e., $|\psi_{k,ij,h}| = T$ for each $k \in [K]$, $i, j \in [d_k]$ and $h \in [d_{-k}]$.

Theorem 1 *Under Assumptions (O1), (M1), (F1), (L1), (E1), (E2) and (R1), for any $k \in [K]$, we have*

$$\frac{1}{d_k} \sum_{j=1}^{d_k} \left\| \widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_{k,j} \mathbf{Q}_{k,j\cdot} \right\|^2 = O_P \left(d_k^{2(\alpha_{k,1} - \alpha_{k,r_k}) - 1} \left(\frac{1}{T d_{-k}} + \frac{1}{d_k} \right) \frac{d^2}{g_s^2} \right) = o_P(1),$$

where g_s is defined in Assumption (R1). Furthermore, for each $i, j \in [d_k]$ and $h \in [d_{-k}]$, if the set $\psi_{k,ij,h}$ defined in (3.3) has cardinality satisfying

$$\frac{|\psi_{k,ij,h}|}{T} = 1 - \eta_{k,ij,h} \geq 1 - \eta = \psi_0,$$

then we have

$$\begin{aligned} \frac{1}{d_k} \sum_{j=1}^{d_k} \left\| \widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k^a \mathbf{Q}_{k,j\cdot} \right\|^2 &= \frac{1}{d_k} \left\| \widehat{\mathbf{Q}}_k - \mathbf{Q}_k \mathbf{H}_k^{a'} \right\|_F^2 \\ &= O_P \left(d_k^{2(\alpha_{k,1} - \alpha_{k,r_k}) - 1} \left[\left(\frac{1}{T d_{-k}} + \frac{1}{d_k} \right) \frac{d^2}{g_s^2} + \min \left(\frac{1}{T}, \frac{\eta^2}{(1 - \eta)^2} \right) \right] \right) = o_P(1). \end{aligned}$$

The proof of the theorem can be found in the supplementary materials of this paper. The two results in Theorem 1 coincide with each other if $\eta = 0$, i.e., there are no missing values. If $\eta = 0$, $K = 2$ and all factors are pervasive, our resulting rate of convergence is $1/(d_k \min(d_k, T d_{-k}))$. This is consistent with, for example, Theorem 1 of Chen and Fan (2023) or Theorem 3.1 of He et al. (2022) (after normalizing like we do). When $K = 1$ with missing data and all factors are pervasive, the rate is $1/(d_1 \min(d_1, T))$, which is the same as the rate in Theorem 1 of Xiong and Pelger (2023) after our normalization. Hence our rate can be considered a generalization to a general order tensor, with general factor strengths and missing data.

We present the two results in the theorem to highlight the difficulty of obtaining consistency when there are missing values. Since a factor loading matrix is not uniquely defined, in the second result in Theorem 1 we are estimating how close $\widehat{\mathbf{Q}}_k$ is to a version of $\mathbf{Q}_k$ in Frobenius norm, namely $\mathbf{Q}_k \mathbf{H}_k^a$, which is still defining the same factor loading space as $\mathbf{Q}_k$ does. With missing data, such a feat is complicated, in the sense that for the j -th row of $\mathbf{Q}_k$, $\widehat{\mathbf{Q}}_{k,j\cdot}$, there corresponds an $\mathbf{H}_{k,j}$ different from $\mathbf{H}_k^a$ in general, so that $\widehat{\mathbf{Q}}_{k,j\cdot}$ is close to $\mathbf{H}_{k,j} \mathbf{Q}_{k,j\cdot}$. The extra rate $\min(1/T, \eta^2/(1 - \eta)^2)$ in the second result is essentially measuring how close each $\mathbf{H}_{k,j}$ is to $\mathbf{H}_k^a$. See Lemma 3 in the supplementary material as well.

Theorem 2 *Under the Assumptions in Theorem 1, suppose we further have $d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k}} = o(d_{-k})$. Define*

$$g_\eta := \min \left(\frac{1}{T}, \frac{\eta^2}{(1 - \eta)^2} \right), \quad g_w := \prod_{k=1}^K d_k^{\alpha_{k,r_k}}.$$

Then we have the following.

1. The error of the estimated factor series has rate

$$\begin{aligned} & \left\| \mathbf{vec}(\widehat{\mathcal{F}}_{Z,t}) - (\mathbf{H}_K^{a'} \otimes \cdots \otimes \mathbf{H}_1^{a'})^{-1} \mathbf{vec}(\mathcal{F}_{Z,t}) \right\|^2 \\ &= O_P \left(\max_{k \in [K]} \left\{ T^{-1} d d_k^{3\alpha_{k,1} - 2\alpha_{k,r_k}} g_s^{-1} + d^2 g_s^{-1} d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} - 1} + g_\eta g_s d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} + 1} \right\} + \frac{d}{g_w} \right). \end{aligned}$$

2. For any $k \in [K]$, $i_k \in [d_k]$, $t \in [T]$, the squared individual imputation error is

$$\begin{aligned} & (\widehat{\mathcal{C}}_{t,i_1,\dots,i_K} - \mathcal{C}_{t,i_1,\dots,i_K})^2 \\ &= O_P \left(\max_{k \in [K]} \left\{ T^{-1} d d_k^{3\alpha_{k,1} - 2\alpha_{k,r_k}} g_s^{-1} g_w^{-1} + d^2 g_s^{-1} g_w^{-1} d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} - 1} + g_\eta g_s g_w^{-1} d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} + 1} \right\} + \frac{d}{g_w^2} \right). \end{aligned}$$

3. The average imputation error is given by

$$\begin{aligned} & \frac{1}{Td} \sum_{t=1}^T \sum_{i_1,\dots,i_K=1}^{d_1,\dots,d_K} (\widehat{\mathcal{C}}_{t,i_1,\dots,i_K} - \mathcal{C}_{t,i_1,\dots,i_K})^2 \\ &= O_P \left(\max_{k \in [K]} \left\{ T^{-1} d_k^{3\alpha_{k,1} - 2\alpha_{k,r_k}} g_s^{-1} + d g_s^{-1} d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} - 1} + d^{-1} g_\eta g_s d_k^{2\alpha_{k,1} - 3\alpha_{k,r_k} + 1} \right\} + \frac{1}{g_w} \right). \end{aligned}$$

The proof can be found in the supplementary materials, which utilizes some rates from the proof of Theorem 3 in the supplementary materials (without the need for extra rate restrictions like Theorem 3 though). If $K \geq 2$, all factors are pervasive and $\eta = 0$ (i.e, no missing values), the squared individual imputation error has rate

$$\max_{k \in [K]} \left(\frac{1}{T d_k} + \frac{1}{d_k^2} \right) + \frac{1}{d} \asymp \frac{1}{\min(T d_1, \dots, T d_K, d_1^2, \dots, d_K^2)}.$$

This rate is the same as the result in Theorem 4 of Chen and Fan (2023) for $K = 2$, which is a rate for estimating the common component. On the other hand, if η is a constant and $K = 1$, and all factors are pervasive, then the second result becomes $d_1^{-1} + T^{-1} \asymp 1/\min(d_1, T)$, which is the same rate as result 3 of Theorem 2 in Xiong and Pelger (2023).

Our generalization has certainly revealed that when there are weak factors, especially when the strongest and weakest factor strengths are quite different, these rates greatly suffer. The complication of missing data comes explicitly from the rate g_η . The average squared imputation error in result 3 improves upon individual squared error in result 2 when weak factors exist, with degree of improvements larger when the difference in strength of factors is larger.

4.3 Inference on the factor loadings

We establish asymptotic normality of the factor loadings for inference purpose. In Section 4.4 we present the covariance matrix estimator for practical use of our asymptotic normality result. Before presenting our results, we need three additional assumptions.

(L2) (Eigenvalues) *For any $k \in [K]$, the eigenvalues of the $r_k \times r_k$ matrix $\Sigma_{A,k} \mathbf{Z}_k$ from Assumption (L1) are distinct.*

(AD1) *There is a constant C such that for any $k \in [K], j \in [d_k]$, as $d_1, \dots, d_K, T \rightarrow \infty$,*

$$\sqrt{\frac{T}{d_{-k} d_k^{\alpha_{k,1}}}} \cdot \left\| \sum_{i=1}^{d_k} \mathbf{H}_k^{a,*} \mathbf{Q}_{k,i} \cdot \sum_{h=1}^{d_k} \frac{1}{|\psi_{k,ij,h}|} \sum_{t \in \psi_{k,ij,h}} \text{mat}_k(\mathcal{E}_t)_{jh} (\mathbf{A}_{-k})'_h \cdot \text{mat}_k(\mathcal{F}_t)' \mathbf{A}_{k,i} \right\| \geq C > 0,$$

where $\mathbf{Q}_k = \mathbf{A}_k \mathbf{Z}_k^{-1/2}$ is the normalised factor loading, $\mathbf{H}_k^{a,*} := \text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k})^{1/2} \cdot \mathbf{D}_k^{-1/2} \Upsilon'_k \mathbf{Z}_k^{1/2}$ with $\mathbf{D}_k := \text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}) \text{diag}\{\lambda_1(\mathbf{A}'_k \mathbf{A}_k), \dots, \lambda_{r_k}(\mathbf{A}'_k \mathbf{A}_k)\}$, and Υ_k the eigenvector matrix of $\text{tr}(\mathbf{A}'_{-k} \mathbf{A}_{-k}) \cdot g_s^{-1} d_k^{\alpha_{k,1} - \alpha_{k,r_k}} \mathbf{Z}_k^{1/2} \Sigma_{A,k} \mathbf{Z}_k^{1/2}$.

(AD2) *Define the filtration $\mathcal{G}^T := \sigma(\cup_{s=1}^T \mathcal{G}_s)$ with $\mathcal{G}_s := \sigma(\{\mathcal{M}_{t,i_1,\dots,i_K} \mid t \leq s\}, \mathbf{A}_1, \dots, \mathbf{A}_K)$, and*

$$\Delta_{F,k,ij,h} := \frac{1}{|\psi_{k,ij,h}|} \sum_{t \in \psi_{k,ij,h}} \text{mat}_k(\mathcal{F}_t) \mathbf{v}_{k,h} \mathbf{v}'_{k,h} \text{mat}_k(\mathcal{F}_t)' - \frac{1}{T} \sum_{t=1}^T \text{mat}_k(\mathcal{F}_t) \mathbf{v}_{k,h} \mathbf{v}'_{k,h} \text{mat}_k(\mathcal{F}_t)',$$

where $\mathbf{v}_{k,h} := [\otimes_{l \in [K] \setminus \{k\}} \mathbf{A}_l]_{h..}$. With $\mathbf{Q}_k$ being the normalised mode- k factor loading in (AD1), we have for every $k \in [K], j \in [d_k]$, for a function $h_{k,j} : \mathbb{R}^{r_k} \rightarrow \mathbb{R}^{r_k \times r_k}$,

$$\begin{aligned} & \sqrt{T d_k^{\alpha_{k,r_k}}} \cdot \mathbf{D}_k^{-1} \mathbf{H}_k^{a,*} \sum_{i=1}^{d_k} \mathbf{Q}_{k,i} \cdot \mathbf{A}'_{k,i} \cdot \sum_{h=1}^{d_k} \Delta_{F,k,ij,h} \mathbf{A}_{k,j} \\ & \rightarrow \mathcal{N}(\mathbf{0}, \mathbf{D}_k^{-1} \mathbf{H}_k^{a,*} h_{k,j}(\mathbf{A}_{k,j \cdot}) (\mathbf{H}_k^{a,*})' \mathbf{D}_k^{-1}) \quad \mathcal{G}^T\text{-stably.} \end{aligned}$$

In (AD1), $\mathbf{H}_k^{a,*}$ is the probability limit of $\mathbf{H}_k^a$ defined in (4.4). Assumption (AD1) is new to the literature. Bai (2003), Chen and Fan (2023) or Xiong and Pelger (2023) all prove asymptotic normality by arguing that in a decomposition, a certain term weakly converging to a normal distribution has asymptotic upper bound dominating the asymptotic upper bounds of other terms. We argue that comparing upper bounds *only* is theoretically not enough to guarantee that the said term's dominance results in asymptotic normality, especially when those upper bounds can be crude and using inequalities like the Cauchy-Schwarz to prove. Assumption (AD1) is to guarantee that the term leading to asymptotic normality is truly dominating others even in the lower bound, which is necessary to guarantee the dominance leading to asymptotic normality. Please refer to the proof of Theorem 3 in the supplementary materials for more details.

Assumption (AD2) is required since the missing data creates a discrepancy term $\Delta_{F,k,ij,h}$ as defined in the assumption. This assumption is also parallel to Assumption G3.5 in Xiong and Pelger (2023). We demonstrate how this assumption is satisfied with Assumption (O1), (F1), (L1) and two additional but simpler assumptions in Proposition 1 in Section 4.6.

Theorem 3 *Let all the assumptions under Theorem 2 hold, in addition to (L2), (AD1) and (AD2) above. With r_k fixed and $d_k, T \rightarrow \infty$ for $k \in [K]$, suppose also $Td_{-k} = o(d_k^{\alpha_{k,1} + \alpha_{k,r_k}})$. We have*

$$\sqrt{Td_k^{\alpha_{k,r_k}}} \cdot (\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k^a \mathbf{Q}_{k,j\cdot}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \mathbf{D}_k^{-1} \mathbf{H}_k^{a,*} (Td_k^{\alpha_{k,r_k}} \cdot \boldsymbol{\Xi}_{k,j} + h_{k,j}(\mathbf{A}_{j\cdot}))(\mathbf{H}_k^{a,*})' \mathbf{D}_k^{-1}), \text{ where}$$

$$\boldsymbol{\Xi}_{k,j} := \text{plim}_{T, d_1, \dots, d_K \rightarrow \infty} \text{Var} \left(\sum_{i=1}^{d_k} \mathbf{Q}_{k,i\cdot} \sum_{h=1}^{d_{-k}} \frac{1}{|\psi_{ij,h}|} \sum_{t \in \psi_{ij,h}} \text{mat}_k(\mathcal{E}_t)_{jh} (\mathbf{A}_{-k})'_h \text{mat}_k(\mathcal{F}_t)' \mathbf{A}_{k,i\cdot} \right).$$

Furthermore, if $Td_{-k}^{-1} g_s^2 g_\eta d_k^{1+\alpha_{k,1}-3\alpha_{k,r_k}} = o(1)$ is also satisfied, then with $\omega_B := d_{-k}^{-1} d_k^{2\alpha_{k,r_k}-3\alpha_{k,1}} g_s^2$,

$$\sqrt{T\omega_B} \cdot (\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k^a \mathbf{Q}_{k,j\cdot}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, T\omega_B \cdot \mathbf{D}_k^{-1} \mathbf{H}_k^{a,*} \boldsymbol{\Xi}_{k,j} (\mathbf{H}_k^{a,*})' \mathbf{D}_k^{-1}).$$

If all factors are pervasive, the rate condition $Td_{-k} = o(d_k^{\alpha_{k,1} + \alpha_{k,r_k}})$ reduces to $Td_{-k} = o(d_k^2)$, which is equivalent to the condition needed for asymptotic normality in Bai (2003) for $K = 1$ and Chen and Fan (2023) for $K = 2$. The first asymptotic normality result is compatible to Theorem 2.1 of Xiong and Pelger (2023) when all factors are pervasive. In their Theorem 2.1, the $\Gamma_{\Lambda,j}^{obs}$ is in fact of rate N^{-1} , so that the normalizing rate is $\sqrt{TN}$, which is exactly $\sqrt{Td_1}$ in our first result when $K = 1$.

Suppose all factors are pervasive. The rate condition $Td_{-k}^{-1} g_s^2 g_\eta d_k^{1+\alpha_{k,1}-3\alpha_{k,r_k}} = o(1)$ is automatically satisfied when there is no missing data, i.e., $\eta = 0$ so that $g_\eta = 0$. If so, the rate of convergence is $\sqrt{T\omega_B} = \sqrt{Td}$, which is compatible to Theorem 2.1 and Theorem 2.2 of Chen and Fan (2023) (after our normalisation to their factor loading matrices). The condition is also satisfied when there is only finite number of missing data, so that $\eta \asymp T^{-1}$ and $g_\eta \asymp T^{-2}$, and $d_1, d_2 = o(T)$ for $K = 2$.

Remark 2 *We do not establish asymptotic normality for the estimated factor series and common components. The reason is that for tensor with $K > 1$, the decomposition in the estimated factor series and the common components cannot be dominated by terms that are asymptotically normal. This is also the reason why Chen and Fan (2023) does not include asymptotic normality for the estimated factor series and common components. We leave the corresponding inference problems in future research.*

4.4 Estimation of covariance matrix

In order to carry out inferences for the factor loadings using Theorem 3, we need to estimate the asymptotic covariance matrix for $\widehat{\mathbf{Q}}_{k,j\cdot} - \mathbf{H}_k^a \mathbf{Q}_{k,j\cdot}$. To this end, we use the heteroscedasticity and autocorrelation

consistent (HAC) estimators (Newey and West, 1987) based on $\{\widehat{\mathbf{Q}}_k, \text{mat}_k(\widehat{\mathcal{C}}_t), \text{mat}_k(\widehat{\mathcal{E}}_t)\}_{t \in [T]}$, where

$$\text{mat}_k(\widehat{\mathcal{C}}_t) = (\widehat{\mathbf{Q}}_k) \text{mat}_k(\widehat{\mathcal{F}}_{Z,t})(\widehat{\mathbf{Q}}_K \otimes \cdots \otimes \widehat{\mathbf{Q}}_{k+1} \otimes \widehat{\mathbf{Q}}_{k-1} \otimes \cdots \otimes \widehat{\mathbf{Q}}_1)', \quad \text{mat}_k(\widehat{\mathcal{E}}_t) := \text{mat}_k(\mathcal{Y}_t) - \text{mat}_k(\widehat{\mathcal{C}}_t).$$

With a tuning parameter β such that $\beta \rightarrow \infty$ and $\beta/(Td_k^{\alpha_k, r_k})^{1/4} \rightarrow 0$, we define two HAC estimators

$$\begin{aligned} \widehat{\Sigma}_{HAC} &:= \mathbf{D}_{k,0,j} + \sum_{\nu=1}^{\beta} \left(1 - \frac{\nu}{1+\beta}\right) (\mathbf{D}_{k,\nu,j} + \mathbf{D}_{k,\nu,j}', \\ \widehat{\Sigma}_{HAC}^{\Delta} &:= \mathbf{D}_{k,0,j}^{\Delta} + \sum_{\nu=1}^{\beta} \left(1 - \frac{\nu}{1+\beta}\right) (\mathbf{D}_{k,\nu,j}^{\Delta} + (\mathbf{D}_{k,\nu,j}^{\Delta})'), \quad \text{where} \\ \mathbf{D}_{k,\nu,j} &:= \sum_{t=1+\nu}^T \left(\sum_{i=1}^{d_k} \left(\frac{1}{T} \sum_{s=1}^T \widehat{\mathbf{D}}_k^{-1} \widehat{\mathbf{Q}}_k' \widehat{\mathbf{C}}_{(k),s} \widehat{\mathbf{C}}_{(k),s,i} \right) \sum_{h=1}^{d_k} \frac{1}{|\psi_{k,ij,h}|} \widehat{\mathbf{E}}_{(k),t,jh} \widehat{\mathbf{C}}_{(k),t,ih} \cdot \mathbb{1}\{t \in \psi_{k,ij,h}\} \right) \\ &\quad \cdot \left(\sum_{i=1}^{d_k} \left(\frac{1}{T} \sum_{s=1}^T \widehat{\mathbf{D}}_k^{-1} \widehat{\mathbf{Q}}_k' \widehat{\mathbf{C}}_{(k),s} \widehat{\mathbf{C}}_{(k),s,i} \right) \sum_{h=1}^{d_k} \frac{1}{|\psi_{k,ij,h}|} \widehat{\mathbf{E}}_{(k),t-\nu,jh} \widehat{\mathbf{C}}_{(k),t-\nu,ih} \cdot \mathbb{1}\{t-\nu \in \psi_{k,ij,h}\} \right)', \\ \mathbf{D}_{k,\nu,j}^{\Delta} &:= \sum_{t=1+\nu}^T \left[\sum_{i=1}^{d_k} \left(\frac{1}{T} \sum_{s=1}^T \widehat{\mathbf{D}}_k^{-1} \widehat{\mathbf{Q}}_k' \widehat{\mathbf{C}}_{(k),s} \widehat{\mathbf{C}}_{(k),s,i} \right) \sum_{h=1}^{d_k} \left(\frac{1}{|\psi_{k,ij,h}|} \widehat{\mathbf{C}}_{(k),t,ih} \widehat{\mathbf{C}}_{(k),t,jh} \cdot \mathbb{1}\{t \in \psi_{k,ij,h}\} \right. \right. \\ &\quad \left. \left. - \frac{1}{T} \widehat{\mathbf{C}}_{(k),t,ih} \widehat{\mathbf{C}}_{(k),t,jh} \right) \right] \cdot \left[\sum_{i=1}^{d_k} \left(\frac{1}{T} \sum_{s=1}^T \widehat{\mathbf{D}}_k^{-1} \widehat{\mathbf{Q}}_k' \widehat{\mathbf{C}}_{(k),s} \widehat{\mathbf{C}}_{(k),s,i} \right) \right. \\ &\quad \left. \cdot \sum_{h=1}^{d_k} \left(\frac{1}{|\psi_{k,ij,h}|} \widehat{\mathbf{C}}_{(k),t-\nu,ih} \widehat{\mathbf{C}}_{(k),t-\nu,jh} \cdot \mathbb{1}\{t-\nu \in \psi_{k,ij,h}\} - \frac{1}{T} \widehat{\mathbf{C}}_{(k),t-\nu,ih} \widehat{\mathbf{C}}_{(k),t-\nu,jh} \right) \right]', \end{aligned}$$

where $\widehat{\mathbf{C}}_{(k),s} := \text{mat}_k(\widehat{\mathcal{C}}_s)$ and $\widehat{\mathbf{E}}_{(k),s} := \text{mat}_k(\widehat{\mathcal{E}}_s)$.

Theorem 4 *Let all the assumptions under Theorem 2 hold, in addition to (L2), (AD1) and (AD2) above. With r_k fixed and $d_k, T \rightarrow \infty$ for $k \in [K]$, suppose also the rate for the individual common component imputation error in result 2 of Theorem 2 is $o(1)$, together with $Td_k = o(d_k^{\alpha_k, 1 + \alpha_k, r_k})$ and $d_k^{2(\alpha_k, 1 - \alpha_k, r_k)} [(Td_k)^{-1} + d_k^{-1}] d^2 g_s^{-2} = o(1)$. Then*

1. $\widehat{\mathbf{D}}_k^{-1} \widehat{\Sigma}_{HAC} \widehat{\mathbf{D}}_k^{-1}$ is consistent for $\mathbf{D}_k^{-1} \mathbf{H}_k^{a,*} \Xi_{k,j} (\mathbf{H}_k^{a,*})' \mathbf{D}_k^{-1}$;
2. $\widehat{\mathbf{D}}_k^{-1} \widehat{\Sigma}_{HAC}^{\Delta} \widehat{\mathbf{D}}_k^{-1}$ is consistent for $(Td_k^{\alpha_k, r_k})^{-1} \mathbf{D}_k^{-1} \mathbf{H}_k^{a,*} h_{k,j} (\mathbf{A}_{k,j}) (\mathbf{H}_k^{a,*})' \mathbf{D}_k^{-1}$;
3. $(\widehat{\Sigma}_{HAC} + \widehat{\Sigma}_{HAC}^{\Delta})^{-1/2} \widehat{\mathbf{D}}_k (\widehat{\mathbf{Q}}_{k,j} - \mathbf{H}_k^a \mathbf{Q}_{k,j}) \xrightarrow{\mathcal{D}} \mathcal{N}(\mathbf{0}, \mathbf{I}_{r_k})$.

The extra rate assumption $d_k^{2(\alpha_k, 1 - \alpha_k, r_k)} [(Td_k)^{-1} + d_k^{-1}] d^2 g_s^{-2} = o(1)$ makes sure that we have Frobenius norm consistency for $\widehat{\mathbf{Q}}_k$ from Theorem 1. The imputation error from result 2 of Theorem 2 also has rate going to 0 when all factors are pervasive, for instance. With result 3 in particular, we can perform inferences on any rows of $\widehat{\mathbf{Q}}_k$. Practical performances of result 3 is demonstrated in Section 5.1.3. The reason that we need two HAC estimators is that similar to Theorem 1, there is a component for missing data, arising from the fact that $\mathbf{H}_{k,j}$ is different from $\mathbf{H}_k^a$ for each $j \in [d_k]$ in general.

4.5 Estimation of number of factors

The reconstructed mode- k sample covariance matrix $\mathbf{S}_k$ is in fact estimating a complete-sample version of a matrix $\mathbf{R}_k^*$, where

$$\mathbf{R}_k^* := \frac{1}{T} \sum_{t=1}^T \mathbf{Q}_k \text{mat}_k(\mathcal{F}_{Z,t}) \mathbf{\Lambda}_k' \mathbf{\Lambda}_k \text{mat}_k(\mathcal{F}_{Z,t})' \mathbf{Q}_k', \quad (4.5)$$

and $\mathcal{F}_{Z,t}$ and $\mathbf{\Lambda}_k$ are defined in (3.2). It turns out that we have $\lambda_j(\hat{\mathbf{S}}_k) \asymp_P \lambda_j(\mathbf{R}_k^*)$ for $j \in [r_k]$, and

$$\lambda_j(\mathbf{R}_k^*) \asymp_P d_k^{\alpha_{k,j} - \alpha_{k,1}} g_s, \quad g_s := \prod_{k=1}^K d_k^{\alpha_{k,1}} \text{ as defined in (R1).}$$

We have the following theorem.

Theorem 5 *Let Assumption (O1), (M1), (F1), (L1), (E1), (E2) and (R1) hold. Moreover, assume*

$$\begin{cases} dg_s^{-1} d_k^{\alpha_{k,1} - \alpha_{k,r_k}} [(Td_{\cdot k})^{-1/2} + d_k^{-1/2}] = o(d_k^{\alpha_{k,j+1} - \alpha_{k,j}}), & j \in [r_k - 1] \text{ with } r_k \geq 2; \\ dg_s^{-1} [(Td_{\cdot k})^{-1/2} + d_k^{-1/2}] = o(1), & r_k = 1. \end{cases}$$

Then $\hat{r}_k$ is a consistent estimator of r_k , where

$$\hat{r}_k := \arg \min_{\ell} \left\{ \frac{\lambda_{\ell+1}(\hat{\mathbf{S}}_k) + \xi}{\lambda_{\ell}(\hat{\mathbf{S}}_k) + \xi}, \ell \in [\lfloor d_k/2 \rfloor] \right\}, \quad \xi \asymp d[(Td_{\cdot k})^{-1/2} + d_k^{-1/2}]. \quad (4.6)$$

The extra rate assumption is satisfied, for instance, when all factors corresponding to $\mathbf{\Lambda}_k$ are pervasive. An eigenvalue-ratio estimator is considered in Lam and Yao (2012) and Ahn and Horenstein (2013), while a perturbed eigenvalue ratio estimator is considered in Pelger (2019). However, all of these estimators are for a vector time series factor model. Our estimator $\hat{r}_k$ in (4.6) extracts eigenvalues from $\hat{\mathbf{S}}_k$, which is not necessarily positive semi-definite. The addition of ξ can make $\hat{\mathbf{S}}_k + \xi \mathbf{I}_{d_k}$ positive semi-definite, while stabilizing the estimator. We naturally assume that $r_k < d_k/2$, which is a very reasonable assumption for all applications of factor models. In fact, our recommended choice of ξ is

$$\xi = \frac{1}{5} d[(Td_{\cdot k})^{-1/2} + d_k^{-1/2}].$$

The requirement that $\xi \asymp d[(Td_{\cdot k})^{-1/2} + d_k^{-1/2}]$ ensures that $\xi = o_P(\lambda_{r_k}(\hat{\mathbf{S}}_k))$ from our rate assumption in the theorem. Our simulations in Section 5.1.2 suggest that this proposal works very well.

4.6 *How Assumption (AD2) can be implied

This section details how Assumption (AD2) can be implied from simpler assumptions. Readers can skip this part and go straight to the next section for a more integral reading experience. We begin by presenting

a proposition.

Proposition 1 *Let Assumption (O1), (F1), (L1) hold. For a given $k \in [K], j \in [d_k]$, assume also the following:*

1. *The mode- k factor is strong enough such that $\alpha_{k,r_k} > 4/5$, and $d_k^{\alpha_{k,1}-\alpha_{k,r_k}} T^{-\epsilon/2} = o(1)$ with some $\epsilon \in (0, 1)$.*
2. *There exists some $\psi_{k,ij}$ such that $\psi_{k,ij} = \psi_{k,ij,h}$ for any $i \in [d_k], h \in [d_{-k}]$. Furthermore, there exists $\omega_{\psi,k,j}$ such that*

$$d_k^{-2} \sum_{i=1}^{d_k} \sum_{l=1}^{d_k} \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,ij}\}}{|\psi_{k,ij}|} - 1 \right) \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,lj}\}}{|\psi_{k,lj}|} - 1 \right) \xrightarrow{p} \omega_{\psi,k,j}.$$

With the above, Assumption (AD2) is satisfied.

Condition 1 and 2 in Proposition 1 are on the factor strength and missingness pattern, respectively. Condition 1 is trivially satisfied if all factors are pervasive. Condition 2 can be easily satisfied by assuming that in $\text{mat}_k(\mathcal{Y}_t)$, all the elements in each row are simultaneously missing with probability $1 - p_0$. We then have

$$\begin{aligned} & d_k^{-2} \sum_{i=1}^{d_k} \sum_{l=1}^{d_k} \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,ij}\}}{|\psi_{k,ij}|} - 1 \right) \left(\frac{T \cdot \mathbb{1}\{t \in \psi_{k,lj}\}}{|\psi_{k,lj}|} - 1 \right) \\ &= d_k^{-2} \sum_{i=1}^{d_k} \sum_{l=1}^{d_k} \left(\frac{T^2 \cdot \mathbb{1}\{t \in \psi_{k,ij}\} \cdot \mathbb{1}\{t \in \psi_{k,lj}\}}{|\psi_{k,ij}| \cdot |\psi_{k,lj}|} - \frac{T \cdot \mathbb{1}\{t \in \psi_{k,ij}\}}{|\psi_{k,ij}|} - \frac{T \cdot \mathbb{1}\{t \in \psi_{k,lj}\}}{|\psi_{k,lj}|} + 1 \right) \\ & \xrightarrow{p} p_0^{-1} - 1, \end{aligned}$$

which is $\omega_{\psi,k,j}$. Similar to Assumption S3.2 in Xiong and Pelger (2023), the value of $\omega_{\psi,k,j}$ can be regarded as a measure of missingness complexity. It is a parameter related to the variance of the stable convergence, and tends to increase when there is a larger portion of data missing.

5 Numerical Results

5.1 Simulation

We demonstrate the empirical performance of our estimators in this section. Note that we do not have comparisons to other imputation methods since to the best of our knowledge, there are no other general imputation methods available for $K > 1$ apart from tensor completion methods for very specific applications as mentioned in the introduction. Under different missing patterns which will be described later, we

investigate the performance of the factor loading matrix estimators, the imputation, and the estimator of the number of factors. We also demonstrate asymptotic normality as described in Theorem 3, followed by an example plot of a statistical power function using result 3 of Theorem 4. Throughout this section, each simulation experiment of a particular setting is repeated 1000 times.

For the data generating process, we use model (3.1) together with Assumption (E1), (E2) and (F1). More precisely, the elements in $\mathcal{F}_t$ are independent standardised AR(5) with AR coefficients 0.7, 0.3, -0.4, 0.2, and -0.1. The elements in $\mathcal{F}_{e,t}$ and ϵ_t are generated similarly, but their AR coefficients are (-0.7, -0.3, -0.4, 0.2, 0.1) and (0.8, 0.4, -0.4, 0.2, -0.1) respectively. The standard deviation of each element in ϵ_t is generated by i.i.d. $|\mathcal{N}(0, 1)|$.

For each $k \in [K]$, each factor loading matrix $\mathbf{A}_k$ is generated independently with $\mathbf{A}_k = \mathbf{U}_k \mathbf{B}_k$, where each entry of $\mathbf{U}_k \in \mathbb{R}^{d_k \times r_k}$ is i.i.d. $\mathcal{N}(0, 1)$, and $\mathbf{B}_k \in \mathbb{R}^{r_k \times r_k}$ is diagonal with the j -th diagonal entry being $d_k^{-\zeta_{k,j}}$, $0 \leq \zeta_{k,j} \leq 0.5$. Pervasive (strong) factors have $\zeta_{k,j} = 0$, while weak factors have $0 < \zeta_{k,j} \leq 0.5$. Each entry of $\mathbf{A}_{e,k} \in \mathbb{R}^{d_k \times r_{e,k}}$ is i.i.d. $\mathcal{N}(0, 1)$, but has independent probability of 0.95 being set exactly to 0. We set $r_{e,k} = 2$ for all $k \in [K]$ throughout the section.

To investigate the performance with missing data, we consider four missing patterns:

- (M-i) Random missing with probability 0.05.
- (M-ii) Random missing with probability 0.3.
- (M-iii) The missing entries have index $(t, i_1, \dots, i_K)$, where

$$0.5T \leq t \leq T, \quad 1 \leq i_k \leq 0.5d_k \text{ for all } k \in [K].$$

- (M-iv) Conditional random missing such that the unit with index j along mode-1 is missing with probability 0.2 if $(\mathbf{A}_1)_{j,1} \geq 0$, and with probability 0.5 if $(\mathbf{A}_1)_{j,1} < 0$.

To test how robust our imputation is under heavy-tailed distribution, we consider two distributions for the innovation process in generating $\mathcal{F}_t$, $\mathcal{F}_{e,t}$ and ϵ_t : 1) i.i.d. $\mathcal{N}(0, 1)$; 2) i.i.d. t_3 .

5.1.1 Accuracy in the factor loading matrix estimators and imputations

For both the factor loading matrix estimators and the imputations, since our procedure for vector time series ($K = 1$) is essentially the same as that in Xiong and Pelger (2023), we show here only the performance for $K = 2, 3$. We use the column space distance

$$\mathcal{D}(\mathbf{Q}, \widehat{\mathbf{Q}}) = \left\| \mathbf{Q}(\mathbf{Q}'\mathbf{Q})^{-1}\mathbf{Q}' - \widehat{\mathbf{Q}}(\widehat{\mathbf{Q}}'\widehat{\mathbf{Q}})^{-1}\widehat{\mathbf{Q}}' \right\|$$

for any given $\mathbf{Q}, \widehat{\mathbf{Q}}$, which is a common used measure in the literature. For measuring the imputation accuracy, we report the relative mean squared errors (MSE) defined by

$$\text{relative MSE}_{\mathcal{S}} = \frac{\sum_{j \in \mathcal{S}} (\widehat{C}_j - C_j)^2}{\sum_{j \in \mathcal{S}} C_j^2}, \quad (5.1)$$

where $\mathcal{S}$ either denotes the set of all missing, all observed, or all available units.

We consider the following simulation settings:

- (Ia) $K = 2, T = 100, d_1 = d_2 = 40, r_1 = 1, r_2 = 2$. All factors are pervasive with $\zeta_{k,j} = 0$ for all k, j . All innovation processes in constructing $\mathcal{F}_t, \mathcal{F}_{e,t}$ and ϵ_t are i.i.d. standard normal, and missing pattern is (M-i).
- (Ib) Same as (Ia), but one factor is weak with $\zeta_{k,1} = 0.2$ for all $k \in [K]$.
- (Ic) Same as (Ia), but all innovation processes are i.i.d. t_3 , and all factors are weak with $\zeta_{k,j} = 0.2$ for all k, j .
- (Id) Same as (Ic), but $T = 200, d_1 = d_2 = 80$.
- (Ie) $K = 3, T = 80, d_1 = d_2 = d_3 = 20, r_1 = r_2 = r_3 = 2$. All factors are pervasive with $\zeta_{k,j} = 0$ for all k, j . All innovation processes in constructing $\mathcal{F}_t, \mathcal{F}_{e,t}$ and ϵ_t are i.i.d. standard normal, and missing pattern is (M-i).
- (If) Same as (Ie), but all factors are weak with $\zeta_{k,j} = 0.2$ for all k, j .
- (Ig) Same as (If), but $T = 200, d_1 = d_2 = d_3 = 40$.

Settings (Ia) to (Id) have $K = 2$, and settings (Ie) to (Ig) have $K = 3$. They all have missing pattern (M-i), but we have considered all settings with missing patterns (M-ii) to (M-iv), with performance of the factor loading matrix estimators very similar to those with missing pattern (M-i). Hence we are only presenting the results for settings (Ia) to (Ig) in Figure 1. The imputation results for the above settings are collected in Table 1, together with those under different missing patterns.

We can see from Figure 1 that the factor loading matrix estimators perform worse when there are weak factors or when the distribution of the innovation processes is fat-tailed. However, larger dimensions ameliorate the worsen performance. The increase in the loading space distance from $k = 1$ to $k = 2$ in settings (Ia) to (Id) is due to more factors along mode-2, which naturally incurs more errors compared to smaller r_k .

From Table 1, we can see that missing pattern (M-iii) is uniformly more difficult in all settings for imputation. This is understandable as there is a large block of data missing in setting (M-iii), so that

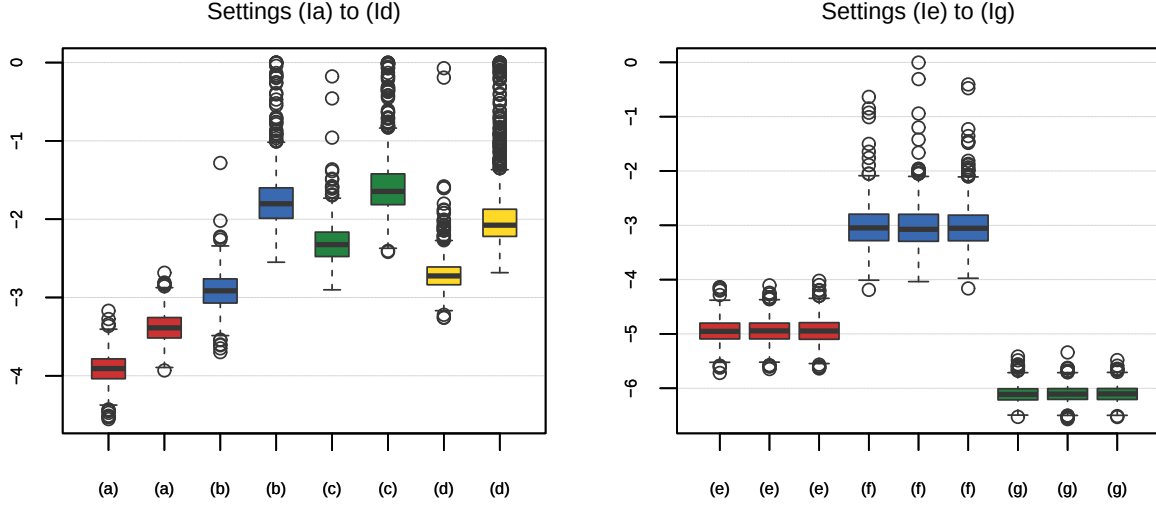


Figure 1: Plot of the column space distance $\mathcal{D}(\mathbf{Q}_k, \widehat{\mathbf{Q}}_k)$ (in log-scale) for $k \in [K]$, with $K = 2$ on the left panel and $K = 3$ on the right. The horizontal axis is indexed by sub-settings (a) to (g), and the k -th boxplot within each sub-setting corresponds to the k -th factor loading matrix $\mathbf{Q}_k$.

we obtain less information towards the “center” of the missing block. This is also the reason why under (M-iii), the imputation performance for the missing set is worse than the observed set, unlike for other missing patterns where all imputation performances are close.

Random missing in (M-i) and (M-ii) are relatively easier for our imputation procedure to handle. Note that if the TALL-WIDE algorithm in Bai and Ng (2021) were to be extended to the case for $K > 1$, it can handle missing pattern (M-iii), but not (M-i) and (M-ii). The design of our method allows us to handle a wider variety of missing patterns, including random missingness. We want to stress that we have made attempts to generalise the TALL-WIDE algorithm to impute high-order time series data for comparisons, yet the method is almost impossible to use in tensor data. The generalisation is also too complicated, and hence is not showed here.

5.1.2 Performance for the estimation of the number of factors

In this section, we demonstrate the performance of our ratio estimator $\widehat{r}_k$ in (4.6) for estimating r_k for $K = 1, 2, 3$. For each $k \in [K]$, we set the value of ξ in Theorem 5 as $\xi = d[(Td_k)^{-1/2} + d_k^{-1/2}]/5$. We have tried a wide range of values other than $1/5$ for ξ in all settings, but $1/5$ is working the best in vast majority of settings, and hence we do not recommend treating this as a tuning parameter for saving computational time.

We present the results under a fully observed scenario and a missing data scenario for each of the following setting:

Setting		K=2				K=3		
Missing Pattern	$\mathcal{S}$	(Ia)	(Ib)	(Ic)	(Id)	(Ie)	(If)	(Ig)
(M-i)	obs	.002	.020	.066	.039	2.61	120	.293
	miss	.002	.020	.066	.039	2.63	121	.294
	all	.002	.020	.066	.039	2.61	120	.293
(M-ii)	obs	.003	.025	.079	.045	5.97	154	.702
	miss	.003	.025	.079	.045	6.06	155	.703
	all	.003	.025	.079	.045	6.00	154	.702
(M-iii)	obs	.004	.025	.079	.048	6.64	136	1.75
	miss	.009	.036	.107	.061	14.7	164	4.02
	all	.005	.026	.083	.050	7.19	138	1.89
(M-iv)	obs	.004	.027	.086	.047	7.75	173	.888
	miss	.004	.028	.088	.047	8.49	179	.964
	all	.004	.027	.086	.047	8.00	175	.914

Table 1: Relative MSE for settings (Ia) to (Ig), reported for $\mathcal{S}$ as the set containing respectively observed (obs), missing (miss), and all (all) units. For $K = 3$, all results presented are multiplied by 10^4 .

(IIa) $K = 1, T = d_1 = 80, r_1 = 2$. All factors are pervasive with $\zeta_{1,j} = 0$ for all j . All innovation processes involved are i.i.d. standard normal. We try missing patterns (M-ii), (M-iii), and (M-iv).

(IIb) Same as (IIa), but one factor is weak with $\zeta_{1,1} = 0.1$.

(IIc) Same as (IIb), but factors are weak with $\zeta_{1,1} = 0.1$ and $\zeta_{1,2} = 0.15$.

(IId) Same as (IIc), but $T = 160$.

(IIIa) $K = 2, T = d_1 = d_2 = 40, r_1 = 2, r_2 = 3$. All factors are pervasive with $\zeta_{k,j} = 0$ for all k, j . All innovation processes involved are i.i.d. standard normal, and the missing pattern is (M-ii).

(IIIb) Same as (IIIa), but all factors are weak with $\zeta_{k,j} = 0.1$ for all k, j .

(IIIc) Same as (IIIb), but $T = d_1 = d_2 = 80$.

(IVa) $K = 3, T = d_1 = d_2 = d_3 = 20, r_1 = 2, r_2 = 3, r_3 = 4$. All factors are pervasive with $\zeta_{k,j} = 0$ for all k, j . All innovation processes involved are i.i.d. standard normal, and the missing pattern is (M-ii).

(IVb) Same as (IVa), but all innovation processes are i.i.d. t_3 .

(IVc) Same as (IVa), but $T = 40$.

Since estimating the number of factors with missing data is new to the literature, it is of interest to explore the accuracy of the estimator under different missing patterns. Hence we explore different missing patterns in setting (IIa). Extensive experiments (not shown here) on the imputation accuracy using misspecified number of factors show that underestimation is harmful, while slight overestimation hardly worsen the performance of the imputations. Thus, for each of the above settings, we also compare

the performance using re-imputation and iTIP-ER by Han et al. (2022), where the re-imputation is done by using both $\hat{r}_k$ and $\hat{r}_k + 1$ to avoid information loss due to underestimating the number of factors, see Table 2 and Table 3.

Setting (IIa) (True $r_1 = 2$)							
Missing Pattern	$\hat{r}$	$\hat{r}_{\text{re},0}$	$\hat{r}_{\text{re},1}$	$\hat{r}_{\text{iTIP, re},0}$	$\hat{r}_{\text{iTIP, re},1}$	$\hat{r}_{\text{full}}$	$\hat{r}_{\text{iTIP, full}}$
	Mean(SD)						
(M-ii)	1.98 _(.13)	1.98 _(.13)	2.00 _(.06)	1.97 _(.18)	1.97 _(.22)	1.99 _(.10)	1.92 _(.28)
(M-iii)	1.92 _(.27)	1.93 _(.26)	1.97 _(.20)	1.90 _(.30)	1.92 _(.31)		
(M-iv)	1.98 _(.14)	1.98 _(.14)	2.01 _(.08)	1.97 _(.17)	1.98 _(.24)		
	Correct Proportion						
(M-ii)	.982	.982	.996	.967	.949	.99	.917
(M-iii)	.921	.93	.96	.901	.898		
(M-iv)	.979	.979	.993	.97	.943		

Table 2: Results for setting (IIa). Each column reports the mean and SD (subscripted, in bracket), followed by the correct proportion of the estimates. The estimator $\hat{r}$ is our proposed estimator; $\hat{r}_{\text{re},0}$ and $\hat{r}_{\text{re},1}$ are similar but used imputed data where the imputation is done using the number of factors as $\hat{r}$ and $\hat{r} + 1$, respectively; $\hat{r}_{\text{iTIP, re},0}$ and $\hat{r}_{\text{iTIP, re},1}$ are iTIP-ER on imputed data (using $\hat{r}$ and $\hat{r} + 1$ respectively); $\hat{r}_{\text{full}}$ and $\hat{r}_{\text{iTIP, full}}$ are our estimator and iTIP-ER on fully observed data (in green), respectively.

Correct Proportion							
Setting	$\hat{r}$	$\hat{r}_{\text{re},0}$	$\hat{r}_{\text{re},1}$	$\hat{r}_{\text{iTIP, re},0}$	$\hat{r}_{\text{iTIP, re},1}$	$\hat{r}_{\text{full}}$	$\hat{r}_{\text{iTIP, full}}$
$K = 1$ (True $r_1 = 1$)							
(IIb)	.556	.556	.886	.526	.765	.633	.53
(IIc)	.626	.626	.762	.594	.668	.67	.539
(IId)	.791	.791	.817	.794	.837	.812	.767
$K = 2$ (True $(r_1, r_2) = (2, 3)$)							
(IIIa)	1	1	1	.995	.995	1	.994
(IIIb)	.978	.978	.987	.985	.989	.981	.986
(IIIc)	.999	.999	1	1	.996	.999	1
$K = 3$ (True $(r_1, r_2, r_3) = (2, 3, 4)$)							
(IVa)	1	1	1	.987	.987	1	.988
(IVb)	.996	.996	.999	.991	.991	1	.991
(IVc)	1	1	1	.999	1	1	1

Table 3: Results for settings (II), (III), and (IV), excluding (IIa). Refer to Table 2 for the definitions of different estimators.

From both Table 2 and 3, it is easy to see that our proposed method generally gives more accurate estimates than iTIP-ER, and it is clear that the re-imputation estimate is at least as good as the initial estimate. In fact, $\hat{r}_{\text{re},1}$ outperforms $\hat{r}_{\text{full}}$ which is based on full observation.

5.1.3 Asymptotic normality

We present the asymptotic normality results for $K = 1, 2, 3$ respectively. For all K considered, we present the result on $(\hat{\mathbf{Q}})_{11}$, with the parameter β of our HAC-type estimator set as $\lfloor \frac{1}{5}(Td_1)^{1/4} \rfloor$. We use (M-i) as the missing pattern for all settings.

When the data is a vector time series ($K = 1$), our approach is similar to Xiong and Pelger (2023), but their proposed estimator includes information of lag 0 only, while we use the HAC-type estimator facilitating more serial information. We do not use a β too large since it proves to be too time consuming to run simulations for high-order high-dimensional data when β is large. Because of this, there is no obvious differences in performances between our HAC estimator and the one in Xiong and Pelger (2023), except that our HAC-type estimator appears to give a smoother plot. Hence we omit the comparisons here.

The data generating process is similar to the ones for assessing the factor loading matrix estimators and imputations, but the parameters are slightly adjusted. All elements in $\mathcal{F}_t$, $\mathcal{F}_{e,t}$, and ϵ_t are now independent standardised AR(1) with AR coefficients 0.05, and we use i.i.d. $\mathcal{N}(0, 1)$ as the innovation process. We stress that we include contemporary and serial dependence among the noise variables through our construction following Assumption (E1) and (E2), while most existing literature demonstrating asymptotic normality display results only for i.i.d. Gaussian noise.

We assume all factors are pervasive in this section. For all $K = 1, 2, 3$, given d_1 , we set $T, d_i = d_1/2$, $i \neq 1$. We generate a two-factor model for $K = 1$, and a one-factor model for $K = 2, 3$. For the settings $(K, d_1) = (1, 1000), (2, 400), (3, 160)$, we consider $(\widehat{\Sigma}_{HAC} + \widehat{\Sigma}_{HAC}^\Delta)^{-1/2} \widehat{\mathbf{D}}_1 (\widehat{\mathbf{Q}}_{1,1} - \mathbf{H}_1^a \mathbf{Q}_{1,1})$. In particular, we plot the histograms of the first and second entry in Figure 2, whereas the corresponding QQ plots are presented in Figure 3.

The plots in Figure 2 provide empirical support to Theorem 3 and result 3 of Theorem 4. For $K = 3$, there are some heavy-tail issues, as seen in the bump at the right tail in the histogram (confirmed by its corresponding QQ plot). The QQ plot for $K = 2$ also hints on this, but the tail is thinned as the dimension increases. Our simulation is similar to that in Chen and Fan (2023) for $K = 2$, but we allow only partial data unobserved and we generalize to any tensor order K . We remark that the convergence rate of the HAC-type estimator is not completely satisfactory, such that relatively large dimension is needed, and it becomes less feasible for some applications. We leave the improvements of the HAC-type estimator to future work.

Lastly, we demonstrate an example of statistical testing for the above one-factor model for $K = 2$. More precisely, we want to test the null hypothesis $\mathcal{H}_0 : \mathbf{Q}_{1,11} = 0$ with a two-sided test. A 5% significance level is used so that we reject the null if $(\widehat{\Sigma}_{HAC} + \Sigma_{HAC}^\Delta)^{-1/2} \widehat{\mathbf{D}}_1 \widehat{\mathbf{Q}}_{1,11}$ is not in $[-1.96, 1.96]$. Each experiment is repeated 400 times and the power function for $\mathbf{Q}_{1,11}$ ranging from -0.02 to 0.02 is presented in Figure 4. The power function is approximately symmetric, and suggests that our test can successfully reject the null if the true value for $\mathbf{Q}_{1,11}$ is away from 0. When $\mathbf{Q}_{1,11} = 0$, the false positive probability is 7.25% which is slightly higher than the designated size of test. This is due to the slow convergence of the HAC estimators, and an increase in dimensions would improve this.

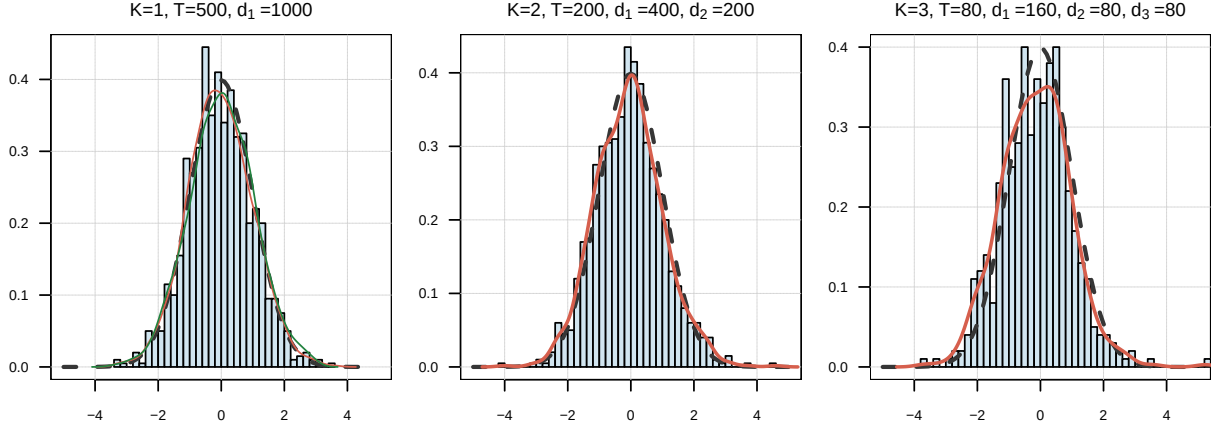


Figure 2: Histograms of the first entry of $(\widehat{\Sigma}_{HAC} + \Sigma_{HAC}^{\Delta})^{-1/2} \widehat{\mathbf{D}}_1(\widehat{\mathbf{Q}}_{1,1} - \mathbf{H}_1^a \mathbf{Q}_{1,1})$. In each panel, the curve (in red) is the empirical density, and the other curve (in green) in the left panel depicts the empirical density of the second entry of $(\widehat{\Sigma}_{HAC} + \Sigma_{HAC}^{\Delta})^{-1/2} \widehat{\mathbf{D}}_1(\widehat{\mathbf{Q}}_{1,1} - \mathbf{H}_1^a \mathbf{Q}_{1,1})$. The density curve for $\mathcal{N}(0, 1)$ (in black, dotted) is also superimposed on each histogram.

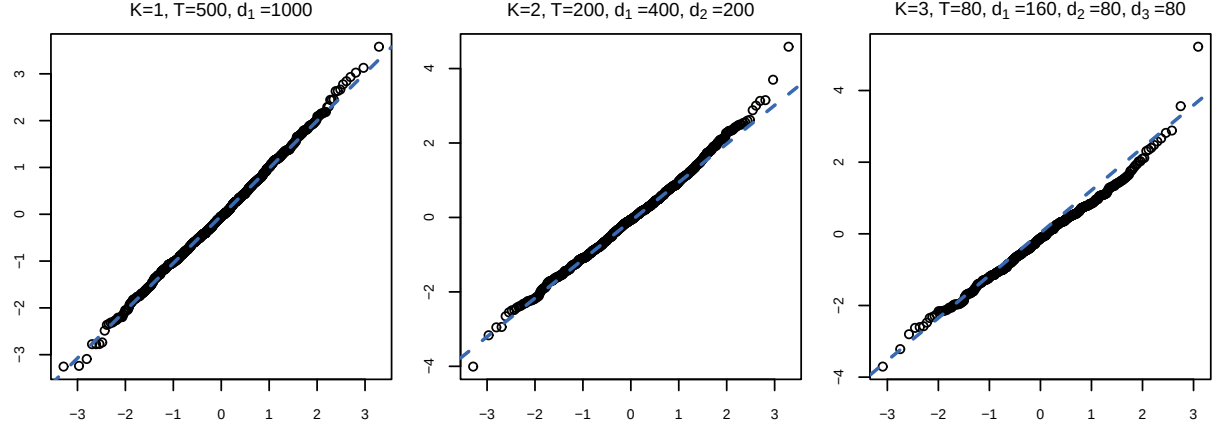


Figure 3: QQ plots of the first entry of $(\widehat{\Sigma}_{HAC} + \Sigma_{HAC}^{\Delta})^{-1/2} \widehat{\mathbf{D}}_1(\widehat{\mathbf{Q}}_{1,1} - \mathbf{H}_1^a \mathbf{Q}_{1,1})$. The horizontal and vertical axes are theoretical and empirical quantiles respectively.

5.2 Real data analysis

5.2.1 Introduction of new measure

We propose a new measure to gauge the imputation performance in this section. Let the observed data be $\mathbf{y} = (y_1, y_2, \dots, y_N)'$ and a vector of imputed data be $\widehat{\mathbf{y}} = (\widehat{y}_1, \widehat{y}_2, \dots, \widehat{y}_N)'$. WLOG, we let the entries in $\mathbf{y}$ be ordered such that $y_1 \leq y_2 \leq \dots \leq y_N$, and each $\widehat{y}_i$ is the corresponding imputed value for y_i . Given $q \in [N]$ and any integer $j = 0, 1, \dots, q$, denote by $q_j \in [N]$ the index of entry with value closest to the (j/q) -th quantile of $\mathbf{y}$. Hereafter, quantiles are always with respect to $\mathbf{y}$ when referred to. For $n \in [q]$,

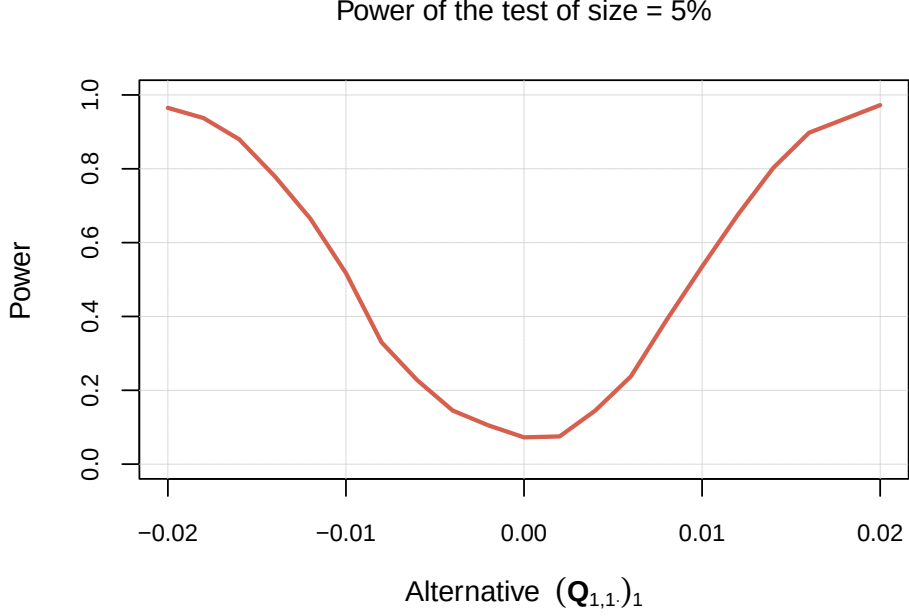


Figure 4: Statistical power of testing the null hypothesis $\mathcal{H}_0 : (\mathbf{Q}_{1,1})_1 = \mathbf{Q}_{1,11} = 0$ against the general alternative. The null is rejected when $|(\widehat{\Sigma}_{HAC} + \Sigma_{HAC}^\Delta)^{-1/2} \widehat{\mathbf{D}}_1 \widehat{\mathbf{Q}}_{1,11}| > 1.96$.

define

$$\mu_n := \frac{1}{q} \sum_{i=q_{n-1}}^{q_n} y_i, \quad \widehat{\mu}_n := \frac{1}{q} \sum_{i=q_{n-1}}^{q_n} \widehat{y}_i.$$

Hence, μ_n is the average of the entries in $\mathbf{y}$ between the $((n-1)/q)$ -th quantile and the (n/q) -th quantile.

With these, we define the q -quantile relative squared error (q -RSE) of $\widehat{\mathbf{y}}$ on $\mathbf{y}$ as

$$q\text{-RSE} := \sum_{n=1}^q (\mu_n - \widehat{\mu}_n)^2 \Big/ \sum_{n=1}^q \mu_n^2 = \sum_{n=1}^q \left\{ \sum_{i=q_n}^{q_{n+1}} (y_i - \widehat{y}_i) \right\}^2 \Big/ \sum_{n=1}^q \left\{ \sum_{i=q_n}^{q_{n+1}} y_i \right\}^2.$$

It is obvious that q -RSE is always non-negative, and q -RSE = 0 if $\widehat{y}_i = y_i$ for all $i \in [N]$. Note that the relative MSE defined in (5.1) is equivalent to q -RSE with $q = N$, and essentially the average within each quantile interval smooths out the idiosyncratic noise in each data point. Intuitively, q -RSE can be considered as a measure of how good the imputation of y_i is, compared with the overall level of y_i in each quantile interval.

As an example to demonstrate the pitfall of relative MSE, consider a one-factor model for vector time series and y_i represents some data entry for $i \in [N]$. Suppose $y_i = c_i + \epsilon_i$ and $\widehat{y}_i = c_i$, where c_i is the

common component to be imputed and ϵ_i is the noise. Hence we have perfect imputation here, and a good measure should have value close to 0. However,

$$\text{relative MSE} = \frac{\sum_{i=1}^N \epsilon_i^2}{\sum_{i=1}^N (c_i + \epsilon_i)^2},$$

which might even be larger than 1 if the noise term is comparable to the common component. For simplicity if we set $q = 1$, then

$$q\text{-RSE} = \frac{(\sum_{i=1}^N \epsilon_i)^2}{(\sum_{i=1}^N c_i + \sum_{i=1}^N \epsilon_i)^2},$$

which goes to 0 if we assume sensibly that as N gets large, $\sum_{i=1}^N \epsilon_i$ goes to 0 at a faster rate than $\sum_{i=1}^N c_i$. Thus, given that we cannot distinguish between the common component and the noise term for a given set of data, the q -RSE is a better measure to gauge imputation performances.

5.2.2 NYC taxi traffic

We analyze a set of taxi traffic data in New York city in this example. The data includes all individual taxi rides operated by Yellow Taxi in New York City, published at

<https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page>.

For simplicity, we only consider the rides within Manhattan Island, which comprises most of the data. The dataset contains 842 million trip records within the period of January 1, 2013 to December 31, 2022. Each trip record includes features such as pick-up and drop-off dates/times, pick-up and drop-off locations, trip distances, itemized fares, rate types, payment types, and driver-reported passenger counts. Our example here focuses on the pick-up and drop-off dates/times, and pick-up and drop-off locations.

To classify the pick-up and drop-off locations in Manhattan, they are coded according to 69 predefined zones in the dataset. Moreover, each day is divided into 24 hourly periods to represent the pick-up and drop-off times each day, with the first hourly period from 0 a.m. to 1 a.m. The total number of rides moving among the zones within each hour are recorded, yielding data $\mathcal{Y}_t \in \mathbb{R}^{69 \times 69 \times 24}$ each day, where $y_{i_1, i_2, i_3, t}$ is the number of trips from zone i_1 to zone i_2 and the pick-up time is within the i_3 -th hourly period on day t . We consider business days and non-business days separately, so that we will analyze two tensor time series. The business-day series is 2,519 days long, while the non-business-day series is 1,133 days long, within the period of January 1, 2013 to December 31, 2022.

As we are more interested in estimation of the tensor factor structure with missingness and hence imputation, we demean the data before imposing different missing patterns, and investigate their effects. See descriptions of missing pattern (M-i) to (M-iii) in Section 5.1. We first estimate the rank of the core tensor using our proposed eigenvalue ratio estimator. Inspired by the simulation results in Section 5.1.2,

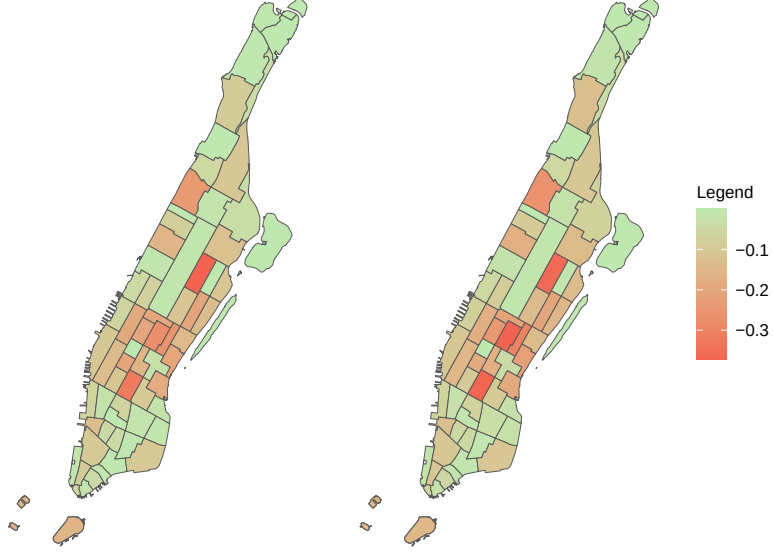


Figure 5: Estimated loading matrices $\hat{\mathbf{A}}_1$ (left) and $\hat{\mathbf{A}}_2$ (right) for business-day series. The data is randomly missing with 5% probability.

we obtain the rank estimates $\hat{r}_k$ for $k \in [K]$, and re-impute the data using $\hat{r}_k + 1$ as the number of factors along mode- k . It appears that the estimated rank is $(1, 1, 1)$ for both the business-day and non-business-day series, and both are consistent under missing patterns (M-i), (M-ii), (M-iii) and no missing. This also coincides with iTIP-ER of Han et al. (2022).

Using the above rank, we estimate the factor loading matrices for different modes. We note the resulting heatmaps of the estimated loadings are almost the same under different missing patterns, so we only display one particular example here. Figure 5 shows the heatmaps of $\hat{\mathbf{A}}_1$ and $\hat{\mathbf{A}}_2$ for the business-day series under missing pattern (M-i). The two heatmaps reveal the same pattern in Manhattan, that traffic is mainly within the downtown areas (the upper east and around the Empire State Building), whereas recreation park or amusement areas (the Central Park and upper west) load slightly on the factor structure. In contrast, dining and sports areas (especially East Harlem) stand out in the factor loadings for non-business-day series. See Figure 6.

The mode-3 loading matrix conveys the information of pick-up time, and the unveiled pattern is the same over different missing patterns. Similar to $\hat{\mathbf{A}}_1$ and $\hat{\mathbf{A}}_2$, we show $\hat{\mathbf{A}}_3$ under missing pattern (M-i) in Table 4 and Table 5 for the business-day and non-business-day series respectively. The highlighted parts indicate the rush-hours in business and non-business days, which depict drastically different patterns. From Table 4, we can see our single factor captures the morning rush-hours (roughly from 9 a.m. to 11 a.m.) and the afternoon-evening rush-hours (between 3 p.m. to 9 p.m.) for business days. Interestingly, we can

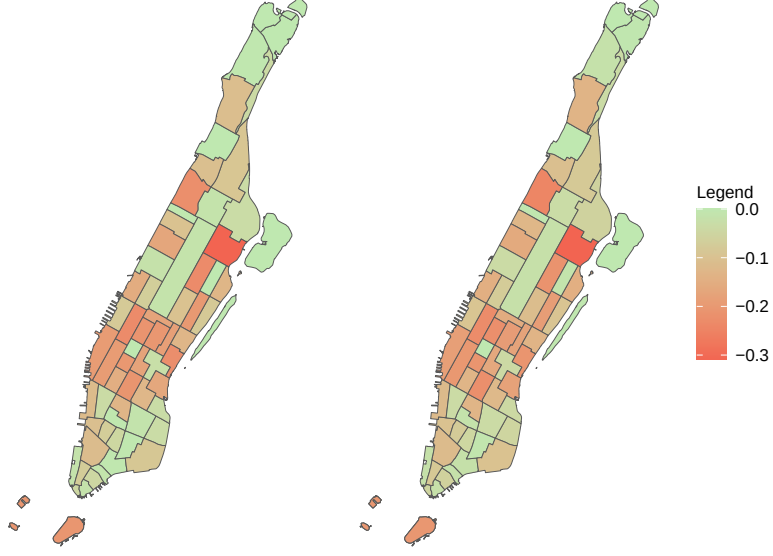


Figure 6: Estimated loading matrices $\hat{\mathbf{A}}_1$ (left) and $\hat{\mathbf{A}}_2$ (right) for non-business-day series. The data is randomly missing with 5% probability.

see the taxi volume spikes slightly at 1 p.m. which might be due to an after-lunch period. We may notice the night life in business days almost end after 1 a.m., while the traffic remains busy even after 2 a.m. in non-business days. It is also clear from the mild sparsity in business-day loading that daily activities are more regular and tractable in business days, compared with those in non-business days.

0 _{am}	12 _{pm}												12 _{am}											
1	3	1	1	0	0	0	1	4	6	6	6	5	6	5	6	6	5	5	6	7	6	5	5	4

Table 4: Estimated loading matrix $\hat{\mathbf{A}}_3$ for business-day series, after scaling. The data is randomly missing with 5% probability. Magnitudes larger than 5 are highlighted in red.

0 _{am}													2	4	6	8	10	12 _{pm}	2	4	6	8	10	12 _{am}
1	5	5	4	3	2	1	1	1	2	3	4	5	5	6	5	5	5	5	6	6	6	5	5	5

Table 5: Estimated loading matrix $\hat{\mathbf{A}}_3$ for non-business-day series, after scaling. The data is randomly missing with 5% probability. Magnitudes larger than 4 are highlighted in red.

From the above discussion, it is easy to see that with our proposed imputation procedure, the tensor factor structure could be well studied even if we cannot observe the full data set. Nevertheless, Table 6 reveals that different missing patterns have slight influence on the imputation accuracy, as also suggested by previous simulation results. The q -RSE are reported for $q=5,10,20,100$, so that the overall performance is gauged accounting for large magnitudes of noise.

Unsurprisingly, we note that the block-missingness in (M-iii) yields the worst accuracy among the experimented patterns. Lastly, the analysis on Manhattan Taxi data is for demonstration of our imputation method, and more sophisticated investigations are required to further understand the traffic pattern.

	business-day			non-business-day		
q	(M-i)	(M-ii)	(M-iii)	(M-i)	(M-ii)	(M-iii)
5	0.279	0.279	0.407	0.289	0.212	0.558
10	0.307	0.307	0.488	0.321	0.271	0.554
20	0.336	0.337	0.515	0.356	0.329	0.541
100	0.377	0.378	0.574	0.417	0.409	0.550

Table 6: q -RSE for business-day and non-business-day series under different missing patterns, with $q = 5, 10, 20, 100$.

6 Appendix

Proofs of all the theorems in this paper can be found in the supplement of this paper at <http://stats.lse.ac.uk/lam/Supp-tensorMiss.pdf>. Instruction in using our R package `tensorMiss` can be found here.

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